

GOPAL MATHUR

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Skills

Languages	Go, Java, C/C++, Python, Rust, TypeScript/JavaScript, Dart, Solidity, Tact
Backend and Frameworks	Spring Boot, Node.js, gRPC, WebSockets, REST APIs, Microservices, Flask
Database	PostgreSQL, SQL, MongoDB, Redis, Vector Databases
Infrastructure & Devops	Docker, Kubernetes, Azure, AWS, CI/CD, Nginx, Kafka, Grafana, Prometheus
AI/ML	Pytorch, Numpy, Pandas, OpenCV, RAG Pipelines, LLM Integration, Transformer Architecture
Frontend	React.js, Next.js, Svelte, Flutter, Slate.js, Android Studio, React Native, Expo

Professional Experience

Siemens Technology and Services

Noida, Uttar Pradesh

Software Development Engineer Intern

May 2025 - Present

- Built a **GenAI PoC** into a production enterprise system, and developed the full-stack architecture from scratch using **Spring Boot**, **React**, and **MongoDB** to support **multi-region adoption across 3 regions** (India, US, and Germany).
- Engineered **RAG pipelines** with **vector embedding-based knowledge** from scratch, to enforce factual accuracy and domain consistency and relevance in generated training material.
- Implemented **token-based rate limiting** on AI generation endpoints to **control API cost exposure**, **prevent abuse**, and fair resource distribution across concurrent users, and maintain stability under load across multi-region usage.
- Containerized** and shipped the full system via **Docker** with **Azure CI/CD pipelines**, owning the deployment architecture end-to-end from local dev to production.

Digital Asset Network

Remote

Smart Contract Engineer Intern

Sep 2024 – Mar 2025

- Executed and deployed production grade **Rust smart contracts** across two blockchain protocols, driving 700+ user adoption with performant on-chain execution.
- Launched functionalities for DeFi financial primitives (lending, rental) in Tact, contributing to **15% customer base growth** by expanding protocol utility surface.
- Authored **TypeScript SDKs** abstracting protocol interactions, enabling third-party developers to integrate financial utilities without direct contract knowledge, reducing integration friction and **expanding ecosystem reach**.
- Optimized **Next.js frontend** with **new protocol integrations** and interaction improvements, resulting in **25% engagement lift**.

Ezinore Private Limited

Rohini, Delhi

Full Stack Developer

Feb 2023 – Sep 2023

- Designed **real-time data pipelines** from ESP32 edge devices to AWS IoT Core in C++, handling **concurrent telemetry ingestion** across 20+ devices with scalable cloud-side message routing
- Built **Flutter** and **SvelteKit prototypes** end-to-end for investor and early adopter demos, owning the full product lifecycle from embedded firmware to user-facing UI

Projects

Realtime Collaborative Document Editor | C++, Go, Java/Spring Boot, React, Kafka, PostgreSQL, Azure AKS, Websockets

- Architected a Google Docs-style **realtime collaborative editor** across polyglot microservices in C++, Go, Java, and React, deployed on **Azure AKS** with **nginx ingress**, cert-manager TLS, and Azure Container Registry.
- Engineered **Operational Transformation** from scratch in **C++** with **Lamport clock-based causal ordering**, exposed via CGo FFI to Go, **handling 400 concurrent ops/sec** and **871 hub churn cycles** under k6 load testing.
- Kafka-based event sourcing** for async communication between Go publishers and Java consumers performing RBAC validation, **idempotency enforcement**, and replayable persistence to **PostgreSQL** across **47,500+ ops** under **sustained load**.
- WebSocket hub engine** in Go with **goroutine-per-connection fan-out**, **panic-safe broadcast**, and **gRPC-based crash recovery**, achieving **738 msgs/sec broadcast throughput** and 100% connection success across **100+ concurrent VUs**.
- RS256 asymmetric JWT** authentication with **refresh token rotation** and JWKS-based key distribution, enabling stateless, zero-network-cost token verification on the hot WebSocket path across microservices.
- Built **text editor frontend** with React and Slate.js, syncing document operations over WebSocket with local OT application for conflict-free multi-user editing.

Custom Decoder-only GPT | Pytorch, Python, Transformer Architecture

- A **GPT-style decoder Transformer Language Model** in **Pytorch** using multi-head attention mechanism, positional embeddings, and causal masking for model training.
- Built the **training pipeline** and **dataset loader** using tiktoken, generating sequence batches for **next-token prediction** and perplexity evaluation.
- Designed and trained a **6-layer Transformer (384-dim, 6 heads, 1024 FFN)** for conversational text completion.

Education

Delhi Technological University

Delhi, India

Bachelor of Technology, Automotive Engineering (CGPA 8.24)

2026

Achievements and Extra-curricular

- President, WEB3DTU (DTU blockchain society) – organized multiple workshops and grew the community to 150+ members.
- Secured 3rd place, Blockbash hackathon (Open category) by Crewsphere ICP Hub India.
- Contributed to E-Summit 2022 website – (E-Cell DTU)

Certifications

Pytorch for Deep Learning - DeepLearning.AI

Spring Boot 3 - LinkedIn Learning

Self Driving Cars Specialization - University of Toronto

Java Object Oriented Programming - LinkedIn Learning